AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 1 · WORKBOOK

The ML Lifecycle

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One connected first-ML-experiment record.

Explain the learning decision, trace a candidate, repair a test-leakage claim and plan the next check.

Keep one habit: Keep task, labels, split, candidate errors, frozen threshold, test predictions and limited conclusion together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Creator Defense. Give an example and a mistake that the idea helps you avoid.

Creator Defense

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 12

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Worked training widths/labels (1,0),(3,0),(5,1),(7,1); candidates 2,4,6 predict 1 when width>=threshold. Mistakes 1,0,1; choose 4 from training only, smaller-threshold tie rule predeclared. Original held-out (2,0),(6,1) predictions 0,1 and 2/2 correct. Pictured plant labels use candidates 2/4 only.

  Separate boundary (4,0) and changed-reference (6,0) probes fail under frozen 4; missing width needs data, not invented 0. Visible reviewed fixtures are practice, not fresh reserved evidence.

- Reference for this exercise: open this week's student lesson, "The ML Lifecycle", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked cards, name the **input**, **label**, three candidate thresholds and the information allowed to select a threshold.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Worked training widths/labels (1,0),(3,0),(5,1),(7,1); candidates 2,4,6 predict 1 when width>=threshold. Mistakes 1,0,1; choose 4 from training only, smaller-threshold tie rule predeclared. Original held-out (2,0),(6,1) predictions 0,1 and 2/2 correct. Pictured plant labels use candidates 2/4 only.

  Separate boundary (4,0) and changed-reference (6,0) probes fail under frozen 4; missing width needs data, not invented 0. Visible reviewed fixtures are practice, not fresh reserved evidence.

- Use threshold 6 on the four **training** widths 1, 3, 5 and 7. Write all **prediction**s, identify each error, and explain why threshold 4 is selected.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Worked training widths/labels (1,0),(3,0),(5,1),(7,1); candidates 2,4,6 predict 1 when width>=threshold. Mistakes 1,0,1; choose 4 from training only, smaller-threshold tie rule predeclared. Original held-out (2,0),(6,1) predictions 0,1 and 2/2 correct. Pictured plant labels use candidates 2/4 only.

  Separate boundary (4,0) and changed-reference (6,0) probes fail under frozen 4; missing width needs data, not invented 0. Visible reviewed fixtures are practice, not fresh reserved evidence.

- Reference for this exercise: open this week's student lesson, "The ML Lifecycle", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- A learner fits on all six worked cards and then reports 6 of 6 as independent test performance. Explain what is wrong and what **evidence** is needed.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Worked training widths/labels (1,0),(3,0),(5,1),(7,1); candidates 2,4,6 predict 1 when width>=threshold. Mistakes 1,0,1; choose 4 from training only, smaller-threshold tie rule predeclared. Original held-out (2,0),(6,1) predictions 0,1 and 2/2 correct. Pictured plant labels use candidates 2/4 only.

  Separate boundary (4,0) and changed-reference (6,0) probes fail under frozen 4; missing width needs data, not invented 0. Visible reviewed fixtures are practice, not fresh reserved evidence.

- After W4, a team changes its threshold because of the test error. Can the same three cards be an untouched final test for the changed model? Explain how to proceed.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use this task's new training cards and candidate thresholds. Record the training mistake counts and selected threshold before checking the three held-out outcomes. Preserve the frozen choice even if a final card fails; a new revision would need a separate evidence record and fresh evaluation.

- New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

  Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

  Give each test **prediction**, the correct count and a limited conclusion.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Fit and evaluate independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Trace one candidate for The **machine learning** lifecycle. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.
- Order the lifecycle and identify missing **evidence**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 1 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

  Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

  Give each test **prediction**, the correct count and a limited conclusion.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Follow one complete plant-classifier lifecycle without tuning on the final test.
- Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/1

The ML Lifecycle

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit using training

Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.





2. Freeze candidate

Choose cutoff 4 under the stated training-error rule.





3. Test once

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

CHANGE ONE THING

The fresh width-6 leaf is actually healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Follow one complete plant-classifier lifecycle without tuning on the final test.
- Which **label**led step matches this detail? Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.
- 1. Test once
- 2. Freeze candidate
- 3. Fit using **training**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Test once Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Read the picture in order.

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The fresh width-6 leaf is actually healthy. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The frozen model is correct on 1/2 test leaves.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The frozen model is correct on 1/2 test leaves. The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Repair and keep the record

Review Creator Defense (ai-3/36). Goal: Explain the **mechanism** and terminology of generative systems defense. Short practice: Which numbers would you display to make the defense honest?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

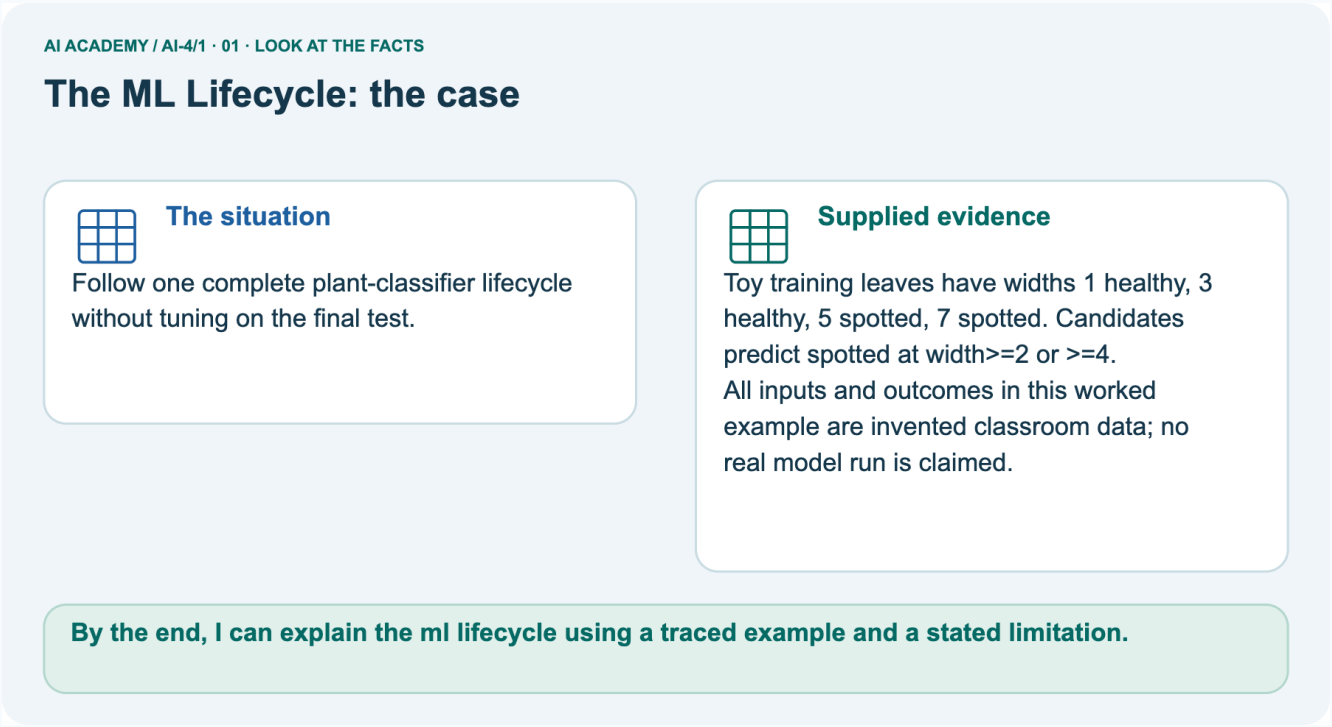
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. The ML Lifecycle: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

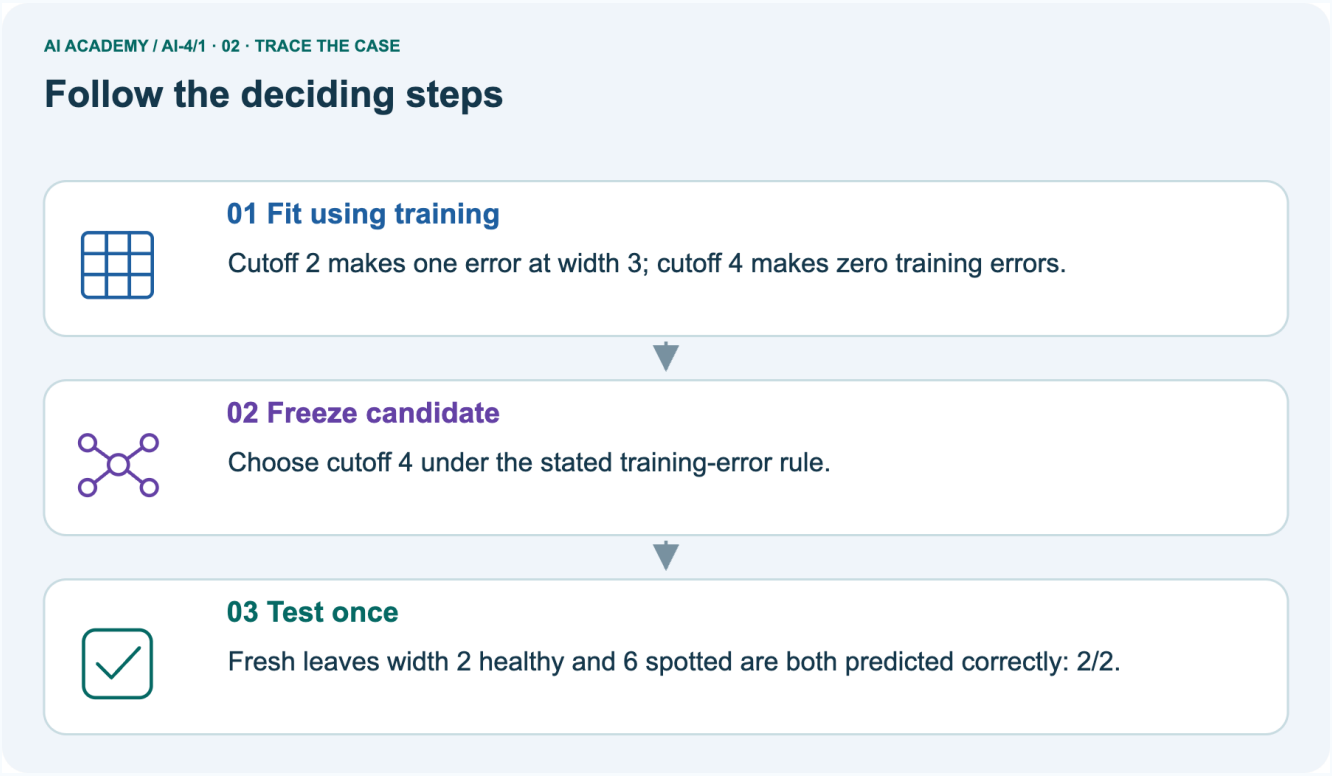
Follow one complete plant-classifier lifecycle without tuning on the final test. Supplied facts: Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



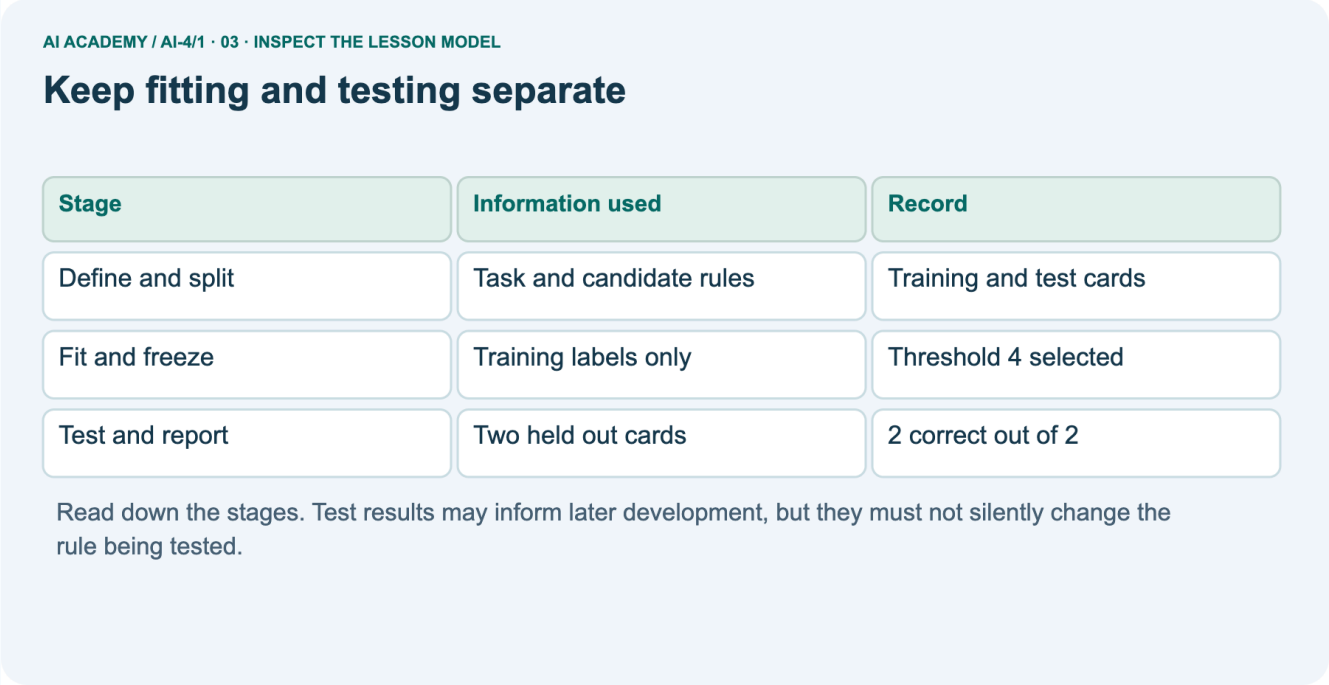
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit using **training**: Cutoff 2 makes one error at width 3; cutoff 4 makes zero **training** errors. → Freeze candidate: Choose cutoff 4 under the stated **training**-error rule. → Test once: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep fitting and testing separate



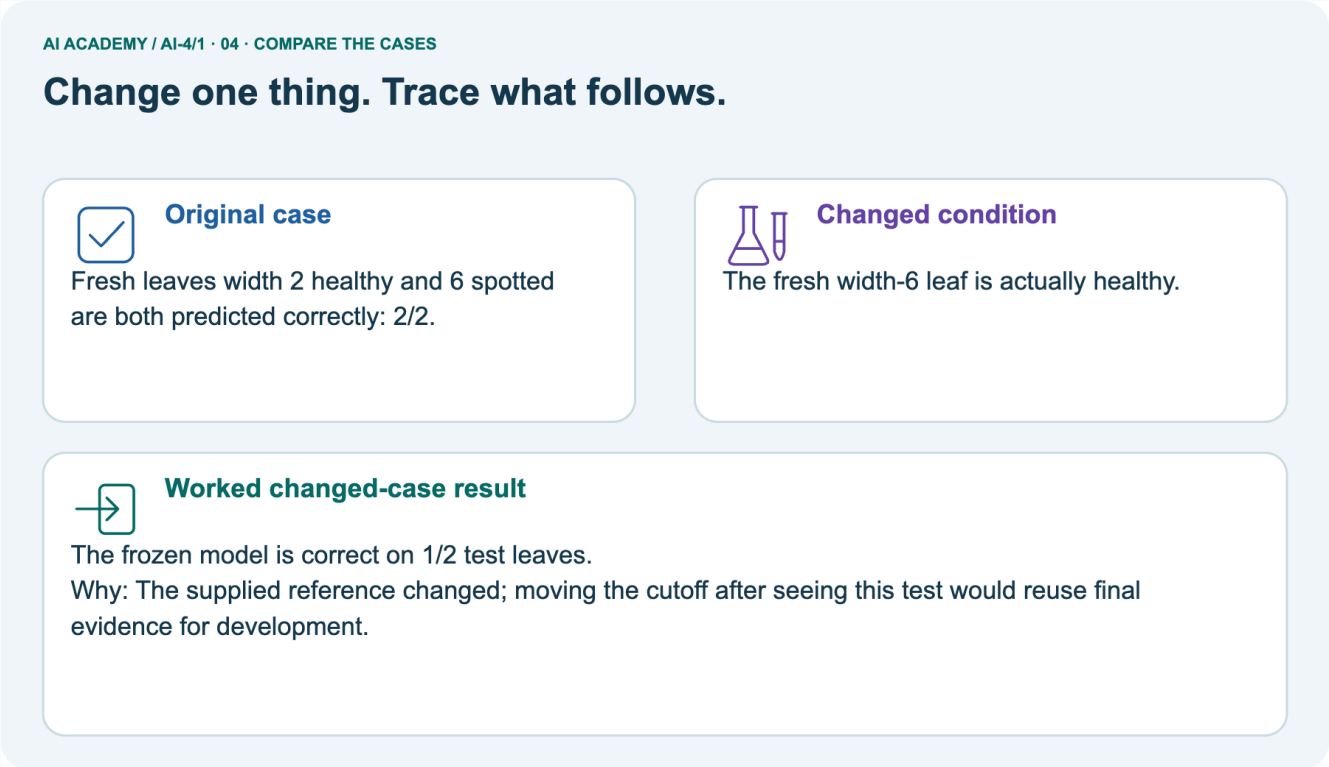
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **fitting** and testing separate. Define and split | Task and candidate rules | **Training** and test cards. Fit and freeze | **Training label**s only | Threshold 4 selected. Test and report | Two held out cards | 2 correct out of 2

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

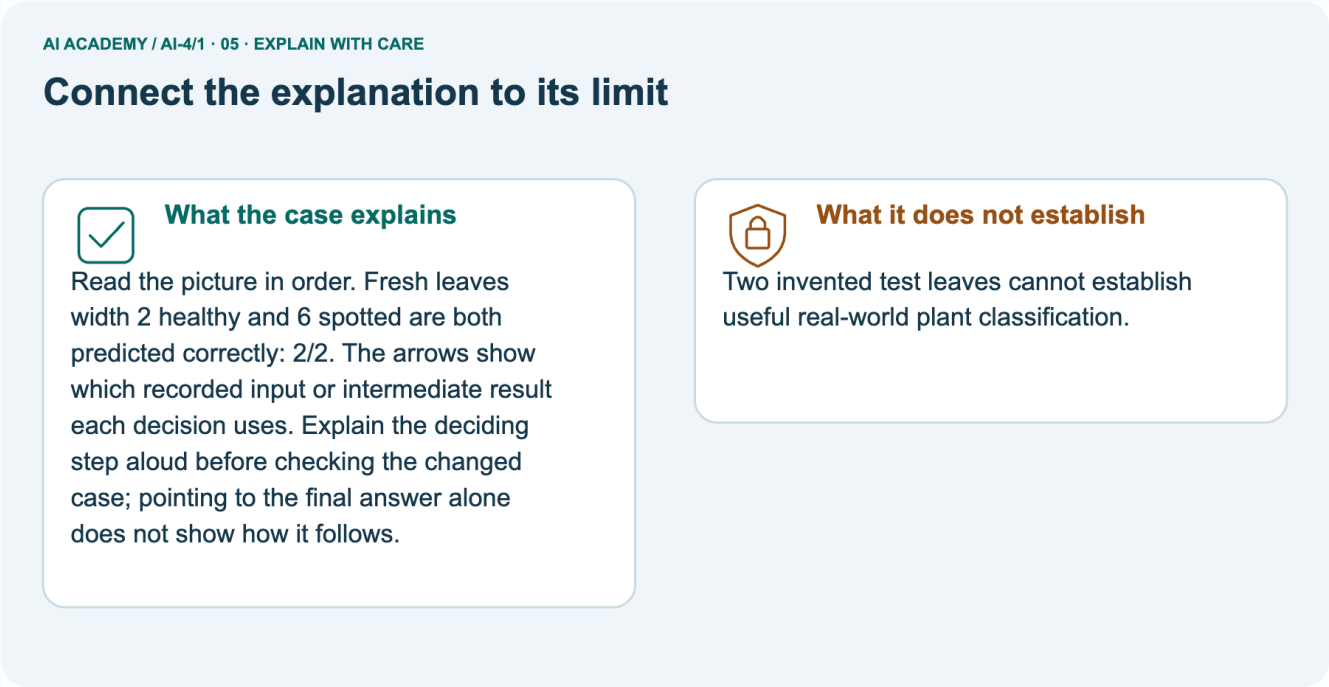
Original case: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Changed condition: The fresh width-6 leaf is actually healthy. Expected result: The frozen model is correct on 1/2 test leaves.

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

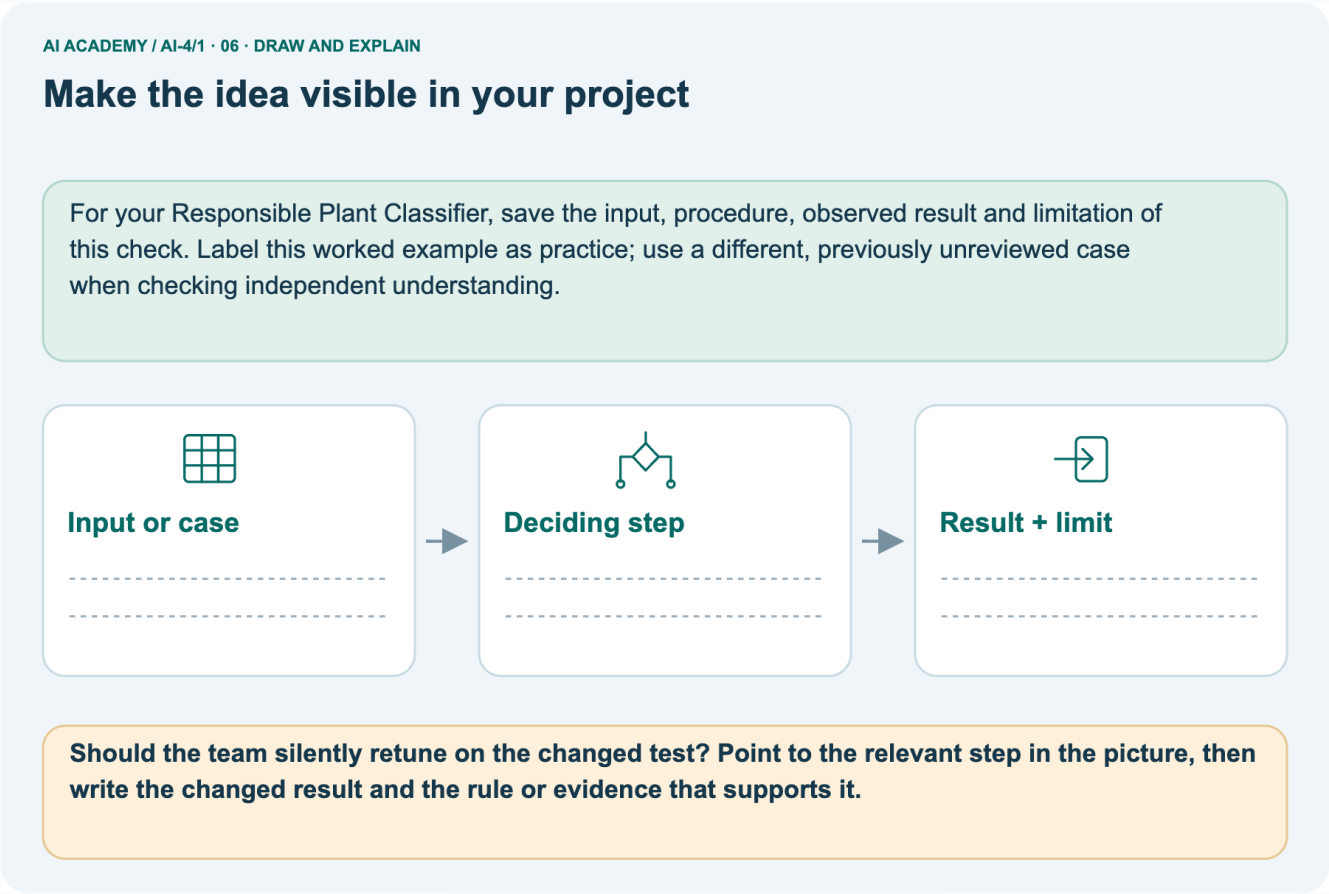
Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Two invented test leaves cannot establish useful real-world plant classification.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. The ML Lifecycle: the case

02. Follow the deciding steps

03. Keep fitting and testing separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.